SANGSAI S T

Software Engineer | Satellite Systems | GNSS/PNT | Scientific Computing

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PROFILE

Final-year Computer Science and Engineering student with hands-on experience developing software for satellite navigation, signal analysis, simulation, and engineering systems. During an internship at VyomIC, contributed to an ITU-based GNSS interference analysis tool, signal modulation and waveform implementations, satellite regulatory data extraction, and Signal-in-the-Loop simulation. Experienced in translating research, mathematical models, and technical requirements into validated Python and MATLAB implementations. Interested in software engineering roles in space, aerospace, navigation, simulation, and deep-tech systems.

CORE COMPETENCIES

Programming: Python, MATLAB, C (Basics)

Scientific & Engineering Software: Signal Processing, GNSS/PNT, Numerical Computing, Simulation, PSD Analysis, Digital Signal Processing

Software & Tools: Git, GitHub, Docker, FastAPI, PostgreSQL, CustomTkinter

Domains: Satellite Systems, LEO-PNT, Interference Analysis, Software-Defined Radio, Sensor Fusion, Digital Twins

EXPERIENCE

GNSS Waveform Design Intern *May 2026 – Present*

VyomIC

- Developed Python-based engineering software for analysing and evaluating satellite navigation signals and system interactions.
- Implemented and validated multiple mathematical and signal-processing models by translating technical literature and reference implementations into production-oriented code.
- Built automation utilities for processing complex engineering and satellite-system datasets to support technical analysis workflows.
- Integrated and validated propagation-delay modelling within a Signal-in-the-Loop simulation environment, addressing discrete-time and signal-preservation challenges.
- Performed debugging, validation, refactoring, documentation, and collaborative development using Git-based workflows.

TECHNICAL PROJECTS

LEO-PNT Cognitive Navigation System | *Python · SDR · IMU · Doppler Fusion · Raspberry Pi*

- Problem: GPS is unavailable in forests, tunnels, urban canyons, and active defense zones — soldiers and rescue teams lose position awareness entirely in these environments.
- Approach: Designed an integrity-aware sensor fusion engine that fuses IMU dead-reckoning with ground-relative radar Doppler velocity aiding and opportunistic LEO satellite Doppler measurements extracted via Software-Defined Radio — no GPS dependency at any layer.
- Impact: Prevents unbounded positional drift without GPS; adaptive source selection means the system degrades gracefully rather than failing catastrophically when one sensor is unavailable.

Landslide Victim Detection System | *Geophone Sensing · ESP32 · Embedded C · Wireless Telemetry*

- Problem: Rescue teams searching debris have no way to detect trapped survivors without physically disturbing unstable rubble, delaying response and risking secondary collapse.
- Approach: Built a passive micro-vibration probe using geophone sensing to detect sub-Hz signals from faint tapping and movement; designed signal processing filters to separate human presence signals from structural debris noise within a 6–7 m radius.
- Impact: Eliminates blind searching; transmits real-time alerts and signal strength wirelessly to surface rescuers, compressing detection time from hours to minutes.

ORCA — Onboard Resource Conflict Arbitrator | *Python · Digital Twin · Object-Oriented Design · Simulation*

- Built a lightweight digital twin for a multi-payload LEO satellite to simulate onboard resource interactions.
- Implemented battery, solar power, payload, power bus, thermal, RF coexistence, telemetry, and spacecraft simulation modules operating at a fixed 1 Hz simulation rate.
- Designed the architecture as a research environment for evaluating future autonomous onboard resource scheduling and conflict arbitration.
- Currently developing the next phase: an onboard decision engine for resource-aware payload scheduling under electrical, thermal, and RF constraints.

Flood Segmentation Pipeline — ANRF AISEHack / *Python · UNet++ · Computer Vision*

- Developed an AI-based flood segmentation pipeline for a national IBM-sponsored challenge.
- Worked on satellite/image-based flood segmentation using a UNet++ architecture.
- Achieved an IoU score of 0.730 and secured 9th position in the competition.

ACHIEVEMENTS

- 1st Place — Idea Presentation, Sona College of Technology (February 2025)
- 1st Place — ELECTROBLITZ '26, 24-Hour Hackathon, KNCET (February 2026)
- 1st Place — Selvam College Hackathon, National Level (March 2026)
- Special Prize (Out of the Box), Sri Ramakrishnan College of Engineering, Coimbatore (March 2026)
- Consolation Prize — MULTIVERSE, National Level 30-Hour Hackathon, M. Kumarasamy College of Engineering (March 2026)
- Finalist — Smart India Hackathon 2025; completed intensive 5-day build phase
- 9th Position — ANRF AISEHack, IBM-sponsored national AI challenge; flood segmentation UNet++ pipeline, IoU 0.730

EDUCATION

B.E. Computer Science & Engineering *2023 – 2027*
Kongunadu College of Engineering & Technology, Trichy, Tamil Nadu